Observability, Memory, and the Dynamics of Cooperation in Heterogeneous Multi-Agent Reinforcement Learning

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Abstract

In this work, we ask which elements of the multi-agent reinforcement learning setting determine whether reciprocal cooperation is learned when agents do not perceive their shared environment identically. Cooperation in social dilemmas matters for biological, human, and artificial agents alike, yet the conditions under which independently learning agents come to cooperate remain contested, and the role of heterogeneous perception is largely unexplored. We use the iterated Prisoner's Dilemma with two-period memory as a testbed and study it with collective reinforcement learning dynamics (CRLD), the deterministic, strategy-average limit of temporal-difference learning. In this limit, the learning process becomes a flow in joint strategy space, and the likelihood of cooperation becomes the size of a basin of attraction. We dissect the setting along three dimensions. First, *observability*: an agent that cannot perceive its own last action cannot implement a reciprocal strategy, and the dynamics collapse to mutual defection, whereas losing sight of the co-player is far less harmful. Second, *memory*: no cooperative equilibrium is reachable with one-period memory; it appears at two periods and its basin grows with longer histories. Third, the *learning rule*: policy-gradient dynamics are bistable between full cooperation and full defection, whereas value-based dynamics converge to a robust interior compromise. Throughout, perceptual asymmetry produces shared defection rather than exploitation. Our findings are relevant for the purposeful design of cooperative algorithms whose agents perceive the world differently.

1 Introduction

Problems of cooperation are ubiquitous. They arise in the evolution of cooperation under natural selection, in human affairs—from climate policy Kersting [2017] and pandemic response Glennerster et al. [2024] to political stalemate Frieden and Lake [2015], Dafoe et al. [2020] and economic distrust Acemoglu and Robinson [2012]—and, increasingly, among intelligent machines Hare and Woods [2020], Bar-Yam [2000]. In a social dilemma, individual incentives and collective welfare pull apart: each actor profits from exploiting the others, or fears being exploited, while the collective does best if all cooperate. Such settings are increasingly modeled as Markov decision processes populated by reinforcement learning (RL) agents—governments, firms, consumers—that adapt their strategies from partial observations of an evolving environment Glennerster et al. [2024], Charpentier et al. [2021].

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Preprint. Under review.

Understanding when self-learning agents come to cooperate *spontaneously*, without being instructed to, is important for three reasons. (1) It offers a route to the emergence of cooperation that does not rely on an evolutionary explanation. (2) It guides the design of learning algorithms that are meant to be cooperative. (3) It gives regulators the background needed to anticipate, and if necessary prevent, undesired collusion among learning agents. While evolutionary theory has catalogued a range of cooperation-promoting mechanisms, from direct reciprocity to spatial structure Schmid et al. [2021], Hilbe et al. [2017], the conditions under which independently learning agents cooperate are still contested, and most of the existing work assumes that all agents perceive their environment in the same way.

Real-world actors, however, do not. A firm and a regulator see different slices of the same market; treaty negotiators hold asymmetric information about each other and about their own past commitments. We therefore ask a focused question: *how does heterogeneity in observability shape whether learning agents come to cooperate?* To answer it cleanly, we isolate observability from all other heterogeneities—learning rate, discount factor, risk attitude—so that what the agents see is the only difference between them.

The difficulty is that multi-agent reinforcement learning is typically highly stochastic and data-inefficient, which makes it hard to tell which features of the setting are decisive. We address this with collective reinforcement learning dynamics (CRLD) Barfuss [2023, 2022], Barfuss et al. [2019], the deterministic limit of temporal-difference learning in which strategy averages replace individual samples. In this limit the learning rule defines a flow in joint strategy space: its fixed points are the equilibria the agents can learn, and, assuming equal likelihood of initial strategies, the size of an equilibrium's basin of attraction indicates how likely it is to be learned. This phase-space perspective lets us move past a single cooperation statistic to the stability landscape that produces it, and to vary observability, memory, and the learning rule on the same footing. We use the iterated Prisoner's Dilemma with two-period memory as our testbed, and check the robustness of the findings across the other canonical social dilemmas, in N-player and spatially extended environments, and in sampled deep RL.

Our contributions are threefold. First, we show that *self-observability* is the load-bearing perceptual channel: an agent that cannot perceive its own last action cannot implement a reciprocal strategy, and the dynamics collapse to mutual defection, whereas masking the co-player's action is far less harmful. Second, we show that the cooperative basin is *created by memory*: it is absent at one-period memory, appears at two periods, and grows as agents condition on longer histories—a deterministic counterpart of the intrinsic-fluctuation route to cooperation Barfuss and Meylahn [2023]. Third, we show that the *learning rule co-determines the attractor*: policy-gradient dynamics are bistable between full cooperation and full defection, whereas value-based dynamics converge to a robust interior compromise. Across all conditions, the dominant failure mode of observational heterogeneity is shared defection, not the exploitation of the less-informed agent by the better-informed one.

2 Methods

2.1 Learning algorithm

We consider two agents, $i \in \{1, 2\}$, each choosing an action $a^i \in \{c, d\}$—a cooperative or a defective act. Their learning is modeled by CRLD Barfuss [2023, 2022], Barfuss et al. [2019]: the deterministic, infinite-memory-batch limit of temporal-difference learning, which replaces stochastic value updates with an exact map in strategy space. We take this machinery as given and recall only what is needed. Each agent's strategy $x^i(a \mid o)$—the probability of action a given observation o—evolves by a replicator–mutator step along its strategy-average temporal-difference error δ^i,

$$x^i_{t+1}(a \mid o) \;=\; \frac{x^i_t(a \mid o)\, \exp\big(\alpha_i\, \beta_i\, \delta^i_t(o, a)\big)}{\sum_b x^i_t(b \mid o)\, \exp\big(\alpha_i\, \beta_i\, \delta^i_t(o, b)\big)}, \tag{1}$$

with learning rate α_i, choice intensity β_i, and discount factor γ_i. The error δ^i is the strategy-average version of the expected, on-policy temporal-difference error—immediate reward plus discounted successor value minus baseline, i.e., the infinite-batch limit of Expected SARSA Sutton and Barto [2018]—whose closed form we take unchanged from Barfuss et al. [2019], Barfuss and Mann [2022].

We study two learning rules within this framework, which differ in whether δ^i updates the strategy directly (policy-gradient, actor–critic) or through state–action values from which the strategy is derived (value-based, SARSA). All hyperparameters can differ between agents; we hold them equal ($\alpha = 0.1$, $\gamma = 0.9$, $\beta = 1$) so that observability is the sole heterogeneity.

Heterogeneous observations. Our modification is to make the agents heterogeneous *observers*. The true state $s \in \mathcal{S}$ is the joint two-round history, but agent i acts on an observation o^i drawn from a *per-agent* observation model $\Omega^i(o^i \mid s) \in \{0, 1\}$ that may alias several true states onto one observation. A full observer uses the identity model ($o^i \equiv s$). A partial observer's Ω^i masks selected coordinates of the most recent round, leaving the agent with the uniform posterior over the states consistent with what it saw,

$$b^i(s \mid o^i) \;=\; \frac{\Omega^i(o^i \mid s)}{\sum_{s'} \Omega^i(o^i \mid s')}, \tag{2}$$

and every strategy average entering δ^i is taken under b^i Barfuss and Mann [2022]. Heterogeneity is precisely the choice of different Ω^i across agents: Agent 1 keeps the full model, while Agent 2's Ω^2 realizes one of the masking regimes of Section 2.3. Concretely, we extend the open-source `pyCRLD` library Barfuss [2023] to carry a separate observation tensor per agent rather than a single shared one.

2.2 Environment

Our focal environment is the iterated Prisoner's Dilemma (IPD), perhaps the most iconic model system for the preconditions of cooperation. Its one-shot payoffs (Table 1) are $R = 1$ for mutual cooperation, temptation $T = 1.2$, sucker's payoff $S = -0.5$, and punishment $P = 0$, satisfying $T > R > P > S$ and $2R > T + S$. With $T > R$ and $P > S$, each agent prefers defection regardless of what the other does; the dilemma is that both would receive more if both cooperated. We use the IPD as a model system to isolate an effect that is likely to persist in larger systems with the same tension between individual incentives and collective welfare.

Table 1: Payoff matrix of the iterated Prisoner's Dilemma (row agent, column agent).

	c	**d**
c	(1, 1)	(-0.5, 1.2)
d	(1.2, -0.5)	(0, 0)

The Prisoner's Dilemma is one of four canonical symmetric two-action social dilemmas. Reordering the same four payoffs yields *Harmony* (cooperation dominant), the *Stag Hunt* (bistable coordination), and *Snowdrift*/*Chicken* (anti-coordination); the exact values are given in Appendix Table 2. We run the full analysis across all four, so as to separate effects driven by the Prisoner's Dilemma's incentive structure from those that hold across dilemmas. Results in the main text refer to the IPD unless stated otherwise.

When the game is repeated, agents can condition their strategies on the actions of past rounds, and mutual defection is no longer inevitable. We embed each stage game into a *two-period memory*: an agent's state is the outcome of the previous two rounds. This is the smallest history that can host the reciprocal strategies—win-stay–lose-shift and tit-for-tat—on which direct reciprocity relies Schmid et al. [2021], Hilbe et al. [2017], yet it remains low-dimensional enough for exhaustive analysis.

These matrix games are also the incentive cores of richer, spatially and temporally extended environments. We study two from the JaxMARL suite Rutherford et al. [2024]: the *coin game*, a spatial Prisoner's Dilemma in which two agents move on a grid and collect own- versus other-colored coins, and the in-the-matrix *arena* (STORM), in which N agents forage colored resources and, on meeting, play a matrix game determined by their inventories. The coin game's incentive core is the Prisoner's Dilemma analyzed here; the arena generalizes it to $N > 2$, letting us ask whether the observability and learning-rule effects survive beyond the dyad.

2.3 Observational heterogeneity

Agent 1 is a *full observer*: it perceives both agents' last two actions without aliasing. Agent 2 is a *partial observer*: along controlled dimensions, parts of the most recent round are masked, aliasing several true states onto one observation (its belief is spread uniformly over them), while the prior round remains fully observed. We study four restrictions of Agent 2; the exact state-to-observation maps are given in Appendix Tables 3–6.

- **Self-aware**: Agent 2 perceives its *own* most recent action but not Agent 1's.
- **Non-self-aware**: Agent 2 perceives Agent 1's most recent action but not its *own*.
- **Cooperation-tracking**: Agent 2 perceives Agent 1's last action only when Agent 1 *cooperated*; Agent 1's defections are masked.
- **Defection-tracking**: Agent 2 perceives Agent 1's last action only when Agent 1 *defected*; Agent 1's cooperations are masked.

2.4 Learning dynamics as a flow

Equation (1) defines a deterministic map on the joint strategy—equivalently, a vector field on strategy space. At every joint strategy, the update points in a definite direction, giving an intuition about where the collective will learn; the fixed points at which the flow vanishes are the learnable equilibria. We probe each setting in two complementary ways. First, we visualize the flow in the plane spanned by the two agents' cooperation probabilities (Figures 1–2), which exposes the stability landscape and, in particular, whether more than one equilibrium is present. Second, we estimate the size of the cooperative basin of attraction by integrating the dynamics from $N = 1{,}200$ random softmax initial strategies to a fixed point (tolerance 10^{-5}, at most $4{,}000$ steps). Since the dynamics are deterministic, the only randomness is in these initial conditions. The *cooperation rate* is the fraction of runs that converge to mutual cooperation (both cooperation probabilities above 0.9): the measure of the cooperative basin under random initialization.

3 Results

Observability. This section shows how what the partial observer perceives shapes the stability landscape. Read as a flow (Figure 1), the heterogeneous IPD exhibits multistability: under full observability a cooperative equilibrium at $(1, 1)$ coexists with a large basin of mutual defection, and only about 14% of random initial strategies converge to cooperation. Symmetric, complete information thus does not make cooperation typical. Degrading the partial observer shrinks the cooperative basin, and *which* coordinate is hidden matters far more than *how much* is hidden. When Agent 2 cannot perceive its own last action (non-self-aware), the flow drains uniformly to defection: an agent that cannot identify its own move cannot "stay" or "shift" relative to it, so the substrate for direct reciprocity is removed. Losing sight of the co-player instead (self-aware) is far milder—a cooperative basin survives—and merely conditioning which of the co-player's moves are visible (cooperation- versus defection-tracking) is close to full observability. Reciprocity is built on self-observability, not on observability of the partner. Across initial conditions the two agents converge to nearly the same strategy, so the failure mode of heterogeneity is *shared* defection rather than exploitation of one agent by the other.

Memory. The cooperative basin is not intrinsic to the game; it is created by memory. Re-running the full-observability dynamics with one-, two-, and four-period memory (Figure 2) shows that with one-period memory the flow does not reach mutual cooperation from any of our initial strategies. A cooperative equilibrium at $(1, 1)$ becomes learnable only at two periods, and its basin enlarges at four. Longer histories allow the agents to represent richer reciprocal strategies (memory-k tit-for-tat and win-stay–lose-shift), which turns cooperation into a deterministic attractor. This is the deterministic counterpart of the finding that the intrinsic fluctuations of finite-batch learning promote cooperation in the one-period-memory IPD Barfuss and Meylahn [2023]: memory and sampling noise enlarge the same cooperative basin by different routes.

Learning rule. Whether cooperation is learnable at all also depends on the learning rule (Appendix Figure 3). Policy-gradient (actor–critic) dynamics are *bistable*: depending on the initial strategies they

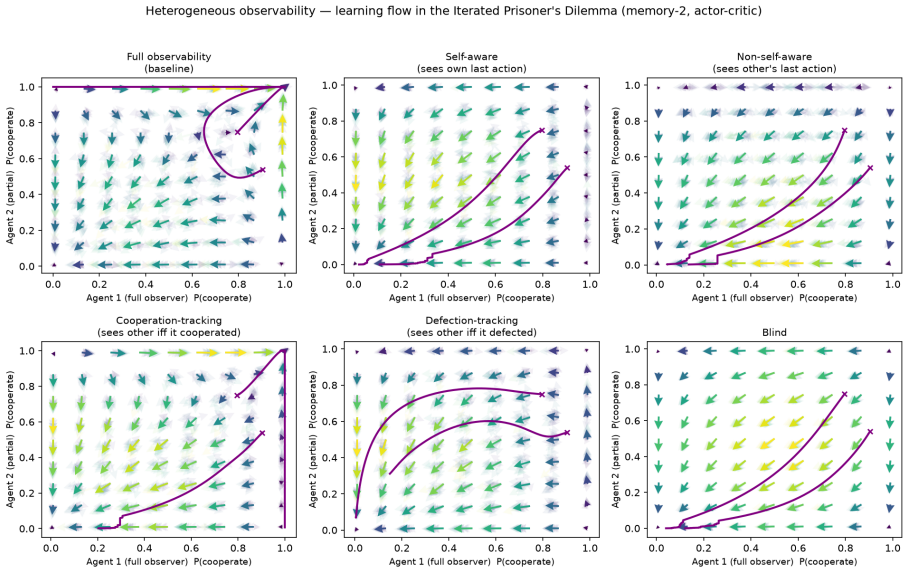


Figure 1: Learning flow of the heterogeneous two-period-memory IPD (actor–critic CRLD), one panel per observability condition of the partial observer (Agent 2; Agent 1 is always a full observer). Arrows show the deterministic learning update at each joint strategy; purple curves are two example trajectories; axes are each agent's cooperation probability. Mutual cooperation is the corner $(1, 1)$, mutual defection $(0, 0)$. A basin draining to cooperation exists under full observability, cooperation-tracking, and—weakly—self-awareness, but the flow collapses toward defection when the partial observer cannot perceive its *own* last action (non-self-aware) or is blind.

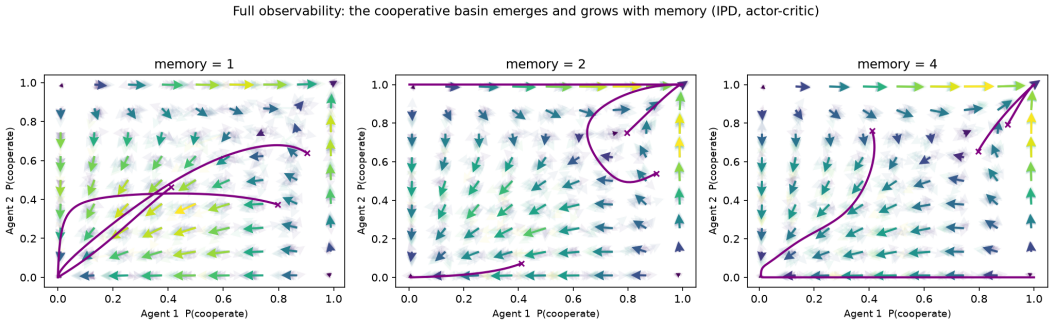


Figure 2: Full-observability IPD learning flow with one-, two-, and four-period memory (actor–critic). A cooperative equilibrium at $(1, 1)$ is not reached with one-period memory, becomes an attractor from two periods onward, and its basin of attraction grows with memory—a deterministic analogue of the intrinsic-fluctuation mechanism Barfuss and Meylahn [2023].

converge either to mutual cooperation $(1, 1)$ or to mutual defection $(0, 0)$, with most random initial conditions in the defection basin. Value-based (SARSA) dynamics instead converge, from almost any initial condition, to a single *interior* partial-cooperation equilibrium: they never collapse to total defection, but never reach full cooperation either. The same game, payoffs, and observability therefore produce qualitatively different cooperative outcomes under the two rules, so the choice of algorithm is part of the cooperation question rather than an implementation detail. In Snowdrift/Chicken the difference is starker still: the two rules select different equilibria outright, a point we return to in Section 4.

Beyond the dyad and the matrix game. The observability and learning-rule effects are not artifacts of the two-player matrix game. In the N-player arena, analyzed in the same deterministic limit, self-observability remains the load-bearing channel (Appendix Figure 4), and cooperation need not

fall as agents are added—in this pairwise-averaged dilemma it can even rise with N. The findings also transfer out of the deterministic limit. The same heterogeneous IPD learned by *sampled* deep RL reproduces the algorithm ordering: value-based IQL climbs to near-full cooperation while policy-gradient IPPO collapses to defection (Appendix Figure 5). The spatial, temporally extended coin game and arena Rutherford et al. [2024] make cooperation harder overall, with a pronounced collapse at large N (Appendix Figure 6).

4 Discussion

Contributions. Reading multi-agent learning as a flow turns the question "does cooperation emerge?" into the sharper question of *what shapes the cooperative basin of attraction*, and we have dissected the answer into three elements. *Observability* sets whether reciprocity can be expressed at all: self-observation is the load-bearing channel, because a reciprocal rule is defined relative to one's own last move, whereas observation of the co-player is comparatively dispensable. *Memory* sets whether a cooperative equilibrium exists: it is absent at one period and is built up as agents condition on longer histories. *The learning rule* sets the shape of the attractor: policy-gradient dynamics are bistable between full cooperation and full defection, whereas value-based dynamics settle on a robust interior compromise. A unifying reading is that cooperation here is gated by how much reciprocal structure the learners can both *represent* and *stabilize*.

Related work. Our results connect three strands. Direct reciprocity is among the best-established mechanisms for the evolution of cooperation Schmid et al. [2021], Hilbe et al. [2017]; we show that it has a perceptual precondition—self-observability—that evolutionary models with fixed strategy spaces take for granted. Deterministic learning dynamics with partial observability have previously been used to show that inaccurate representations of the environment can be beneficial for cooperation Barfuss and Mann [2022]; our contribution is to make the observation model *heterogeneous* across agents and to identify which coordinates of the observation matter. Finally, the intrinsic fluctuations of finite-batch temporal-difference learning are known to double the rate of cooperation in the one-period-memory IPD Barfuss and Meylahn [2023]. Memory and such fluctuations expand the same basin by different routes—more representable history versus noise that escapes the defection trap—which suggests they are partly interchangeable; whether fluctuations can rescue the non-self-aware case that memory alone cannot is a natural test. Deep MARL in sequential social dilemmas Rutherford et al. [2024] provides the high-dimensional end of the bridge that our low-dimensional analysis anchors.

Limitations and future work. The result we find most provocative, and least understood, is *algorithm-driven equilibrium selection*. In Snowdrift/Chicken, value-based dynamics reliably settle on the symmetric, mutually cooperative mixed equilibrium, whereas policy-gradient dynamics are drawn to the asymmetric anti-coordination outcomes: one learning rule discovers a "good" cooperative solution that the other simply does not find. We have no mechanistic account of *why* a learning rule prefers one equilibrium over another, nor of which rules induce cooperative selection in which games. A predictive theory of this would matter well beyond our setting—it would tell a designer which learner to deploy to obtain a desired social outcome—and we regard it as the most promising thread to pull next. Beyond it, natural steps are to (i) sweep β and γ to map where observational heterogeneity becomes cooperation-promoting; (ii) reintroduce finite-batch fluctuations Barfuss and Meylahn [2023] to test whether they rescue the non-self-aware case; and (iii) push the coin game and arena Rutherford et al. [2024] further, and on to language-model agents. The present conclusions are correspondingly bounded. The matrix games capture a narrow slice of strategic behavior; CRLD describes idealized, deterministic learners in the infinite-batch limit rather than human idiosyncrasy; the outcomes depend on the parameters and regimes we report; and a cooperation rate is a statement about the measure of a basin of attraction under random initial strategies, not about any single trajectory.

Practical implications. Two implications follow for the design of cooperative learning systems whose agents perceive the world differently. First, an agent's access to its *own* past actions is not a convenience but a precondition for reciprocity: designers who prune an agent's observation space should preserve self-monitoring before anything else, and should give agents enough history to represent conditional strategies. Second, the learning algorithm is itself a lever on the social

outcome; the same environment can yield bistable all-or-nothing cooperation or a stable partial compromise depending on whether a policy-gradient or a value-based rule is deployed. For regulators, the dominant failure mode is instructive: perceptual asymmetry does not let the better-informed agent dominate but costs both agents the cooperative surplus, so the welfare loss from informational heterogeneity is shared rather than redistributed.

5 Acknowledgments

This project was supported by Impact Academy and the University of Bonn. The authors declare no conflict of interest.

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A Appendix / supplemental material

A.1 Software and reproducibility

All experiments use an extension of the open-source `pyCRLD` library Barfuss [2023] that adds heterogeneous per-agent observation tensors. The deterministic dynamics, observation-tensor construction, and figure-generation scripts for every result and figure in this paper are released with the camera-ready version.

Relation to an earlier preliminary report. An earlier exploratory write-up by the authors reported that knowledge-asymmetric agents cooperate readily and that better-informed agents earn more. Those conclusions do not survive a controlled analysis: the optimistic reading came partly from a run with an out-of-range discount factor ($\gamma = 10$, which makes the $(1-\gamma)$ prefactor negative and the value solve ill-posed) and from per-scenario reporting drift, and the apparent reward advantage of the better-informed agent vanishes—indeed reverses—once a valid $\gamma = 0.9$ is used. All results here use one consistent, flow-based pipeline at $\gamma = 0.9$.

A.2 Parameters

For all experiments we use learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$, and choice intensity $\beta = 1$, identical across both agents. Memory length is two ($h = (2, 2, 2)$ in the history-embedding convention) except in the memory sweep (Figure 2), which varies it. Each scenario uses $N = 1{,}200$ random softmax initial strategies; trajectories run up to $4{,}000$ steps with fixed-point tolerance 10^{-5}.

A.3 Game family

The four canonical symmetric two-action social dilemmas share the structure of Table 1 and differ only in the ordering of R, T, S, P (Table 2). This ordering determines the equilibrium structure—cooperation-dominant (Harmony), a single defective dominant strategy (Prisoner's Dilemma), bistable coordination (Stag Hunt), or anti-coordination (Snowdrift/Chicken)—and lets us test which observability, memory, and learning-rule effects are specific to the Prisoner's Dilemma.

Table 2: Payoffs of the four social dilemmas (mutual cooperation R, temptation T, sucker S, punishment P) and their defining inequality.

Game	R	T	S	P	ordering
Prisoner's Dilemma	1	1.2	-0.5	0	$T > R > P > S$
Harmony	1	0.5	0.5	0	$R > T,\ S > P$
Stag Hunt	1	0.5	-0.5	0	$R > T > P > S$
Snowdrift / Chicken	1	1.5	0.5	0	$T > R > S > P$

A.4 Learning rule

A.5 Richer environments: N-player and deep RL

The claim that the effects survive beyond the two-player matrix game rests on the following supplementary analyses. Figure 4 repeats the observability flow grid for the N-player arena ($N = 3$, deterministic CRLD): self-awareness remains the load-bearing channel, as in the dyad. Figure 5 leaves the deterministic limit and trains sampled deep-RL agents on the same heterogeneous-observation IPD; the value-based learner (IQL) reaches near-full cooperation while the policy-gradient learners (IPPO, A2C) decay toward defection, reproducing the deterministic algorithm ordering. Figure 6 shows the spatial, temporally extended STORM arena Rutherford et al. [2024], where cooperation is harder to reach and collapses as the number of agents grows to $N = 8$.

A.6 Observation maps

Tables 3–6 give the state-to-observation maps for Agent 2 in each scenario. Each row is one of the 16 two-period-memory states; entries show what Agent 2 perceives, with $*$ denoting a masked

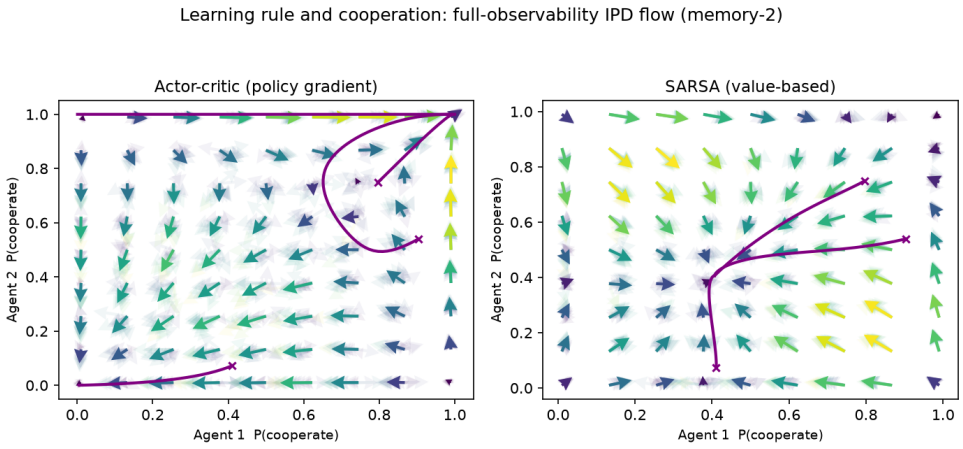


Figure 3: The learning rule reshapes the attractor (full-observability IPD, two-period memory). *Left:* policy-gradient (actor–critic) dynamics are bistable—initial conditions decide between mutual cooperation $(1, 1)$ and mutual defection $(0, 0)$. *Right:* value-based (SARSA) dynamics converge from almost any initial condition to a single interior partial-cooperation equilibrium, avoiding total defection but never reaching full cooperation.

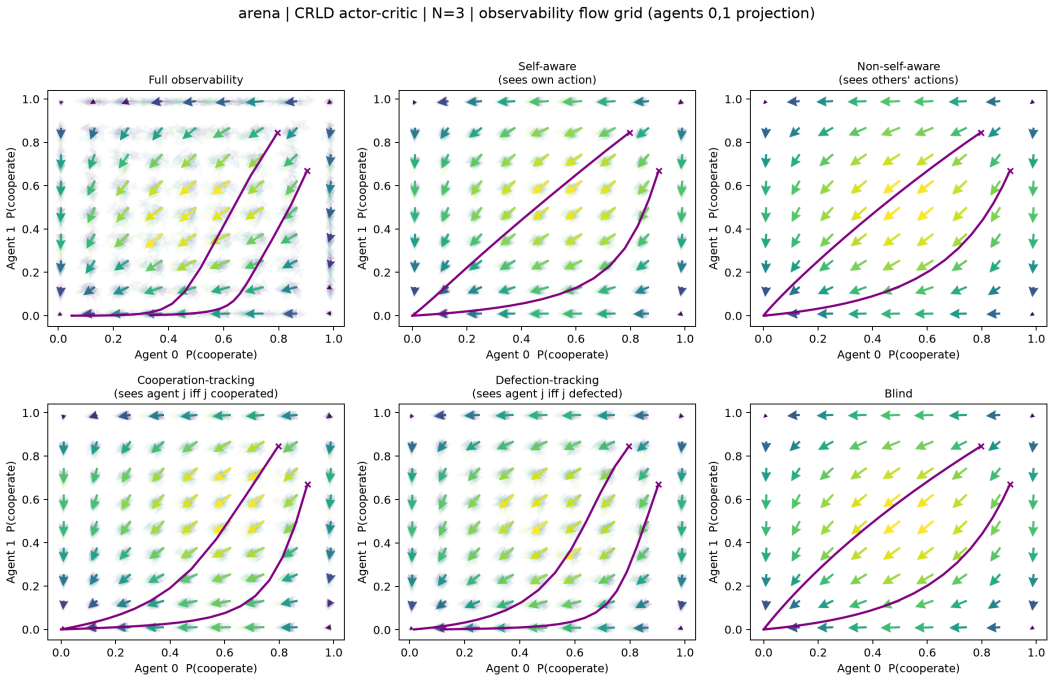


Figure 4: N-player arena ($N = 3$), deterministic CRLD observability flow grid (actor–critic). As in the two-player IPD, self-awareness (top-middle) preserves a cooperative basin while non-self-awareness (top-right) and blindness drain toward defection—the observability finding is not specific to the dyad.

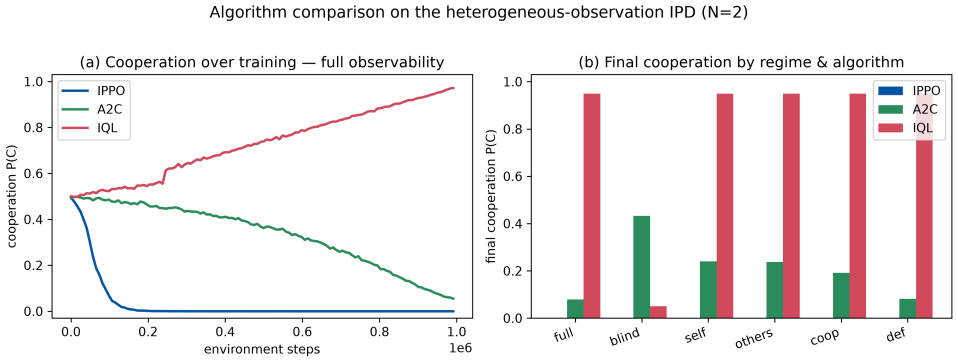


Figure 5: Sampled deep RL on the heterogeneous-observation IPD (away from the deterministic limit). (a) Cooperation over training under full observability: value-based IQL rises toward full cooperation, policy-gradient IPPO collapses to defection, A2C decays slowly. (b) Final cooperation by observability regime and algorithm: the value-based/policy-gradient ordering holds across regimes, reproducing under sampling the algorithm dependence found in the deterministic flow.

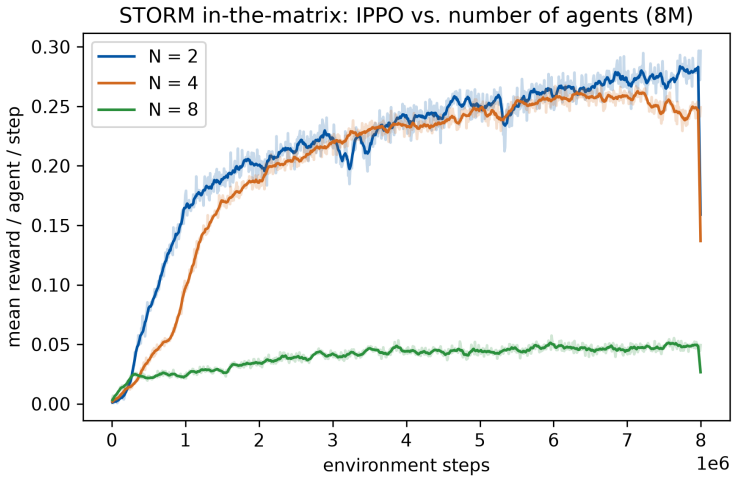


Figure 6: STORM in-the-matrix arena Rutherford et al. [2024] (deep RL, IPPO): mean reward per agent over training for $N = 2, 4, 8$ agents. Cooperation is harder than in the dyadic matrix game and collapses at $N = 8$—an effect of the spatial, temporally extended setting that the deterministic dyadic analysis does not capture.

(unobserved) coordinate that aliases the state with others sharing the same observation. Agent 1 observes every state without aliasing in all scenarios.

Table 3: Self-aware Agent 2: perceives its own most recent action, Agent 1's is masked.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(∗, C)
(C, C)	(C, D)	(C, C)	(∗, D)
(C, D)	(C, C)	(C, D)	(∗, C)
(C, D)	(C, D)	(C, D)	(∗, D)
(D, C)	(D, C)	(D, C)	(∗, C)
(D, C)	(D, D)	(D, C)	(∗, D)
(D, D)	(D, C)	(D, D)	(∗, C)
(D, D)	(D, D)	(D, D)	(∗, D)

Table 4: Non-self-aware Agent 2: perceives Agent 1's most recent action, its own is masked.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, ∗)
(C, C)	(C, D)	(C, C)	(C, ∗)
(C, D)	(C, C)	(C, D)	(C, ∗)
(C, D)	(C, D)	(C, D)	(C, ∗)
(D, C)	(D, C)	(D, C)	(D, ∗)
(D, C)	(D, D)	(D, C)	(D, ∗)
(D, D)	(D, C)	(D, D)	(D, ∗)
(D, D)	(D, D)	(D, D)	(D, ∗)

Table 5: Cooperation-tracking Agent 2: perceives Agent 1's last action only when Agent 1 cooperated.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(C, C)
(C, C)	(C, D)	(C, C)	(C, D)
(C, C)	(D, C)	(C, C)	(∗, C)
(C, C)	(D, D)	(C, C)	(∗, D)
(D, C)	(C, C)	(D, C)	(C, C)
(D, C)	(C, D)	(D, C)	(C, D)
(D, C)	(D, C)	(D, C)	(∗, C)
(D, C)	(D, D)	(D, C)	(∗, D)

Table 6: Defection-tracking Agent 2: perceives Agent 1's last action only when Agent 1 defected.

Agent 1, Game 1	Agent 2, Game 1	Agent 1, Game 2	Agent 2, Game 2
(C, C)	(C, C)	(C, C)	(∗, C)
(C, C)	(C, D)	(C, C)	(∗, D)
(C, C)	(D, C)	(C, C)	(D, C)
(C, C)	(D, D)	(C, C)	(D, D)
(D, C)	(C, C)	(D, C)	(∗, C)
(D, C)	(C, D)	(D, C)	(∗, D)
(D, C)	(D, C)	(D, C)	(D, C)
(D, C)	(D, D)	(D, C)	(D, D)